VI Tutorial

MA(1) and AR(1): moments, autocorrelations and persistence

Let ε_t be i.i.d. over integer time, with $E[\varepsilon_t] = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2 > 0$. Write $\gamma_h = \mathrm{Cov}(y_t, y_{t-h})$ and $\rho_h = \gamma_h/\gamma_0$ for a weakly stationary process. The intercept c and model coefficients are fixed constants.

Exercise 1. Moving average: MA(1)

Consider

$$y_t = c + \varepsilon_t + \theta\varepsilon_{t-1}.$$

Derive the unconditional mean, variance and autocovariances, and hence the ACF for $h \geq 0$. Explain which innovations are shared between two observations at lag one and why the autocovariance is zero at higher lags.

Compare the theoretical ACFs for $\theta \in \{-0.75, -0.5, 0, 0.5, 0.75\}$ in the left panel below. Discuss the sign of lag-one dependence, the case $\theta = 0$ and how long a single innovation directly affects the process.

Exercise 2. Autoregression: AR(1)

Consider

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \qquad t \geq 1.$$

Start from a fixed initial value y_0, with $\phi \in \mathbb{R}$ unrestricted. Do not assume stationarity.

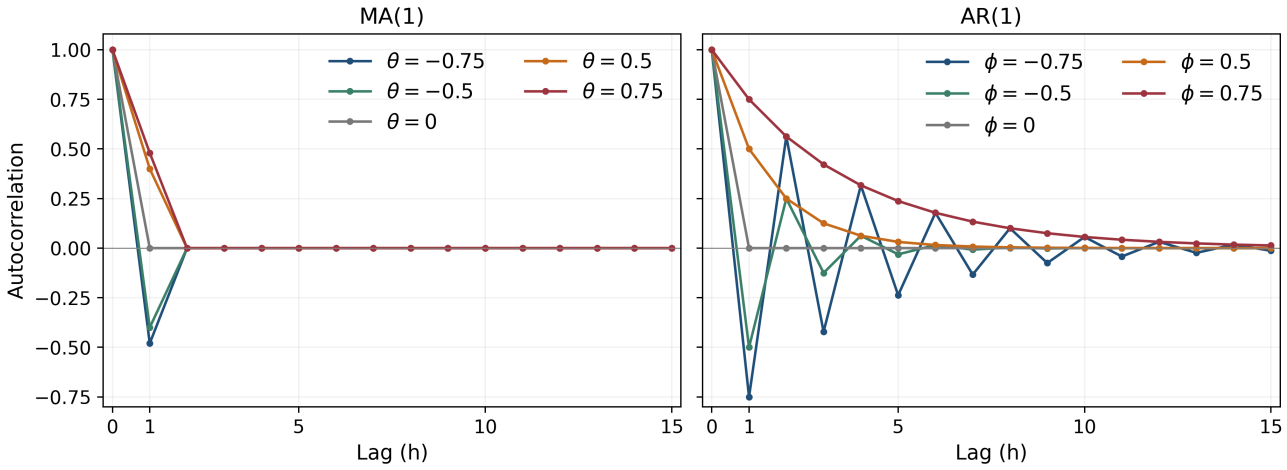
(a) Use repeated substitution to express y_t in terms of y_0, c and the innovations from dates 1 to t.

(b) Derive $E[y_t]$, $\mathrm{Var}(y_t)$ and $\mathrm{Cov}(y_t, y_{t-h})$ for $0 \leq h \leq t$. Treat $\phi = 1$ and $\phi = -1$ separately where needed.

(c) Study the limits as $t \to \infty$ for $|\phi| < 1$, $|\phi| = 1$ and $|\phi| > 1$. Distinguish convergence to stationary moments from exact weak stationarity. If y_0 is instead random and independent of future innovations, what moments give exact weak stationarity? Derive the ACF of that stationary process.

Hint: In the AR(1) recursion, the accumulated intercept is $c\sum_{j=0}^{t-1}\phi^j = c(1-\phi^t)/(1-\phi)$ *for* $\phi \neq 1$, *and* ct *for* $\phi = 1$.

Compare the theoretical ACFs for $\phi \in \{-0.75, -0.5, 0, 0.5, 0.75\}$ in the right panel. Discuss positive and negative coefficients, the speed of decay and the case $\phi = 0$. Contrast the AR(1)'s persistence with the MA(1)'s finite dependence.



Theoretical population ACFs, including $\rho_0 = 1$. The panels use the same vertical scale; there are no sample confidence bands.

Solutions

Exercise 1. Moving average: MA(1)

Mean and variance

The constant shifts the mean but does not affect variance. Independence makes the covariance between distinct innovations zero:

$$\begin{aligned} E[y_t] &= c + E[\varepsilon_t] + \theta E[\varepsilon_{t-1}] = c, \\ \gamma_0 &= \operatorname{Var}(\varepsilon_t + \theta\varepsilon_{t-1}) \\ &= \sigma^2 + \theta^2\sigma^2 + 2\theta \operatorname{Cov}(\varepsilon_t, \varepsilon_{t-1}) = \sigma^2(1+\theta^2). \end{aligned}$$

Autocovariances

Centre the process: $x_t = y_t - c = \varepsilon_t + \theta\varepsilon_{t-1}$. Then $x_{t-1} = \varepsilon_{t-1} + \theta\varepsilon_{t-2}$, so

$$\begin{aligned} \gamma_1 &= E[x_t x_{t-1}] \\ &= E[\varepsilon_t\varepsilon_{t-1}] + \theta E[\varepsilon_t\varepsilon_{t-2}] + \theta E[\varepsilon_{t-1}^2] + \theta^2 E[\varepsilon_{t-1}\varepsilon_{t-2}] \\ &= 0 + 0 + \theta\sigma^2 + 0 = \theta\sigma^2. \end{aligned}$$

Only the shared innovation ε_{t-1} contributes. For $h \geq 2$, x_t and x_{t-h} contain no shared innovations, giving $\gamma_h = 0$. The mean and variance are finite and constant, and autocovariance depends only on the lag: the process is weakly stationary for every finite θ.

ACF and interpretation

Dividing each autocovariance by γ_0 gives

$$\rho_h = \frac{\gamma_h}{\gamma_0} = \begin{cases} 1, & h = 0, \\ \theta/(1+\theta^2), & h = 1, \\ 0, & h \geq 2. \end{cases}$$

The ACF cuts off after lag one. Its lag-one value has the sign of θ; for $\theta = \pm 0.5$ it is ± 0.4, and for $\theta = \pm 0.75$ it is ± 0.48. At $\theta = 0$, $y_t - c$ is white noise. A one-unit innovation affects y_t by 1 and y_{t+1} by θ, with no direct effect thereafter.

Exercise 2. Autoregression: AR(1)

(a) Repeated substitution

Starting from y_0, the first substitutions give

$$\begin{aligned} y_1 &= c + \phi y_0 + \varepsilon_1, \\ y_2 &= c + \phi(c + \phi y_0 + \varepsilon_1) + \varepsilon_2 \\ &= c(1+\phi) + \phi^2 y_0 + \phi\varepsilon_1 + \varepsilon_2. \end{aligned}$$

Continuing the same pattern gives, for every real ϕ,

$$\boxed{y_t = \phi^t y_0 + c\sum_{j=0}^{t-1}\phi^j + \sum_{j=0}^{t-1}\phi^j \varepsilon_{t-j}.}$$

The terms represent the initial value, accumulated intercept and weighted innovations. No stationarity assumption has been used.

(b) Moments at a finite date

Since y_0 is fixed and the innovations have zero mean,

$$E[y_t] = \phi^t y_0 + c\sum_{j=0}^{t-1}\phi^j = \begin{cases} \phi^t y_0 + c\dfrac{1-\phi^t}{1-\phi}, & \phi \neq 1, \\ y_0 + ct, & \phi = 1. \end{cases}$$

Independence makes all covariances between distinct innovations zero. The initial value and intercept contribute no variance, so

$$\begin{aligned} V_t := \mathrm{Var}(y_t) &= \mathrm{Var}\left(\sum_{j=0}^{t-1}\phi^j\varepsilon_{t-j}\right) = \sigma^2\sum_{j=0}^{t-1}\phi^{2j} \\ &= \begin{cases} \sigma^2\dfrac{1-\phi^{2t}}{1-\phi^2}, & |\phi| \neq 1, \\ t\sigma^2, & |\phi| = 1. \end{cases} \end{aligned}$$

Here $V_0 = 0$. The finite geometric sum, not a constant-variance assumption, produces these expressions.

Iterating back h dates, for $1 \le h \le t$, gives

$$y_t = c\sum_{j=0}^{h-1}\phi^j + \phi^h y_{t-h} + \sum_{j=0}^{h-1}\phi^j\varepsilon_{t-j}.$$

The last sum contains innovations after $t-h$, independent of y_{t-h}. Taking covariance with y_{t-h} therefore gives

$$\boxed{\mathrm{Cov}(y_t, y_{t-h}) = \phi^h V_{t-h}}, \qquad 0 \le h \le t.$$

For $h = 0$ this is V_t; for $h = t$ it is zero because y_0 is fixed. In general the covariance depends on the date as well as the lag.

Exercise 2. Autoregression: AR(1) (continued)

(c) Long-run behaviour and stationarity

Stable case, $|\phi| < 1$. Since $\phi^t \to 0$ and $\phi^{2t} \to 0$, the finite-date formulas give

$$E[y_t] \to \mu = \frac{c}{1-\phi}, \qquad V_t \to \gamma_0 = \frac{\sigma^2}{1-\phi^2}, \qquad \mathrm{Cov}(y_t, y_{t-h}) \to \phi^h \gamma_0$$

for each fixed lag h. These are **limits of the moments**; they do not make the fixed-start process exactly stationary from $t = 0$. Even setting $y_0 = \mu$ only fixes its mean: the variance still builds up from $V_0 = 0$.

Boundary cases, $|\phi| = 1$. At $\phi = 1$ the model is a random walk with drift: $E[y_t] = y_0 + ct$ and $V_t = t\sigma^2$. At $\phi = -1$,

$$E[y_t] = \frac{c}{2} + (-1)^t \left(y_0 - \frac{c}{2}\right), \qquad V_t = t\sigma^2.$$

The variance diverges in both cases, even when the mean is constant.

Explosive case, $|\phi| > 1$. The variance is

$$V_t = \sigma^2 \frac{\phi^{2t} - 1}{\phi^2 - 1} \to \infty.$$

Thus only $|\phi| < 1$ produces finite stationary moment limits from this fixed initial value. We have obtained this range by following the process forward, rather than assuming equal moments at successive dates.

Exact weak stationarity and the ACF

Now allow a random y_0, independent of future innovations, with mean m_0 and finite variance v_0. Its contribution gives

$$E[y_t] = \phi^t m_0 + c \sum_{j=0}^{t-1} \phi^j, \qquad \mathrm{Var}(y_t) = \phi^{2t} v_0 + \sigma^2 \sum_{j=0}^{t-1} \phi^{2j}.$$

For $|\phi| < 1$, choose $m_0 = \mu = c/(1-\phi)$ and $v_0 = \gamma_0 = \sigma^2/(1-\phi^2)$. Substituting gives, for every $t \geq 0$,

$$E[y_t] = \phi^t \mu + \mu(1 - \phi^t) = \mu,$$
$$\mathrm{Var}(y_t) = \phi^{2t} \gamma_0 + \gamma_0 (1 - \phi^{2t}) = \gamma_0.$$

The covariance argument in (b) still applies, giving

$$\gamma_h = \phi^h \gamma_0, \qquad \boxed{\rho_h = \frac{\gamma_h}{\gamma_0} = \phi^h}, \qquad h \geq 0.$$

The moments are now constant and autocovariance depends only on the lag: the process is **exactly weakly stationary**. Conversely, with an initial value independent of future innovations, constant finite variance would require $(1 - \phi^2)\gamma_0 = \sigma^2 > 0$, which rules out $|\phi| \geq 1$.

Interpreting persistence

The right panel shows this **stationary ACF**. Positive ϕ gives positive decay; negative ϕ gives alternating signs. Larger $|\phi|$ means slower decay; $\phi = 0$ gives white noise about c. A one-unit innovation has response ϕ^h after h periods, whereas an MA(1) innovation has no direct effect beyond the next period.